ModelIt!

Teacher's Guide

Does the Sun Warm Everything the Same?

Sunlight, Shade, and Temperature

Quick Reference

Grade Level	Kindergarten
NGSS Standard	K-PS3-1
Time	45–60 minutes (core lesson)
Materials	Student Activity Pack, devices with ModelIt! access
Prep	Print activity packs, test ModelIt! on student devices
Grouping	Partners or small groups (2–4)

ModelIt! — Discovery Collective

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1 NGSS Standards & CAST Alignment

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Performance Expectation: K-PS3-1

Make observations to determine the effect of sunlight on Earth's surface.

Three-Dimensional Learning

Science Practice	**Planning and Carrying Out Investigations** — Students feel and compare temperatures of sunny versus shady surfaces, collecting simple observational data.
Disciplinary Core Idea	**PS3.B Conservation of Energy and Energy Transfer** — Sunlight warms Earth's surface. The warmth felt from sunlight is energy that has been transferred from the sun to objects on Earth.
Crosscutting Concept	**Cause and Effect** — Students observe that more sunlight (cause) leads to higher surface temperature (effect).

CAST Testing Alignment

2 Scoring Rubric

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Assessment Reference

This is the same rubric students see in their Activity Pack. Use it during and after the lesson for formative and summative assessment. Circle or highlight the proficiency level for each skill.

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

Scoring Guidance

Expert (4): Student demonstrates deep understanding and can transfer knowledge to new contexts.
Proficient (3): Student meets grade-level expectations and shows solid understanding.
Developing (2): Student is making progress but needs additional scaffolding or practice.
Beginning (1): Student needs significant support; consider using Differentiation strategies (Section 7).

3 Background Content Knowledge

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For Your Confidence

This section provides the science background you need to facilitate this lesson with authority. You don't need to teach all of this — it's here so you can answer student questions accurately.

The sun is the primary source of energy for Earth's surface. Understanding how sunlight warms different areas helps students make sense of daily experiences like seeking shade on a hot day or noticing that metal playground equipment gets hot in the sun.

How Sunlight Warms the Earth

Sunlight is a form of energy called radiant energy. When sunlight hits a surface — soil, concrete, water, or any material — the light energy is absorbed and converted to thermal energy (heat). Darker surfaces absorb more light and get hotter, while lighter surfaces reflect more light and stay cooler. This is why a black car parked in the sun feels much hotter than a white car.

The Role of Shade

Shade is created when an object blocks sunlight from reaching a surface. Trees, buildings, umbrellas, and clouds all create shade. In shaded areas, surfaces receive less direct sunlight and therefore absorb less energy, staying cooler. Shade does not make things cold — it simply reduces the amount of solar energy reaching the surface. On a 90-degree day, shaded ground might be 20-30 degrees cooler than sunlit ground.

4 The LEVER Framework

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About LEVER

LEVER is how every ModelIt! lesson works. Students move through five phases to build, test, and refine a computational model. Each lesson is self-contained — students complete all five phases in one session.

L — **Locate the System**
Students identify Amount of Sunlight as the external component (nature controls whether a spot is sunny or shady) and Surface Temperature as the internal component (it changes based on how much sun a spot gets).

E — **Establish Relationships**
Students discover that more sunlight leads to higher Surface Temperature — a positive relationship.

V — **Visualize & Model**
Students build a simple two-part model connecting Amount of Sunlight to Surface Temperature with a positive arrow.

E — **Evaluate Outcomes**
Students test Shady Spot vs. Sunny Spot and compare how warm each feels.

R — **Revise & Extend**
Students explore how different materials (dark paper vs. white paper) heat up differently in the sun.

Dropping ModelIt! Into Your Existing Unit

ModelIt! is a **supplement**, not a replacement for your science curriculum. Each lesson stands alone, but it becomes even more powerful when students revisit their model throughout a unit. Here are easy ways to weave it in:

At the start of a unit	Use this lesson to introduce the phenomenon. Students build a “first draft” model based on what they already know. Save it.
After a reading or lab	Have students reopen their model and update it: “Based on what you just learned, would you add a component? Change a relationship?” Takes 10–15 minutes.
As a “what-if” tool	When students ask questions during class, say: “Great question — go test that in your model!” Students run new scenarios and report back.
For peer discussion	Partners compare models: “Why does your model have this arrow and mine doesn’t?” Differences spark scientific debate using evidence.
As a presentation tool	Students project their model and walk the class through it live — showing components, running scenarios, answering audience “what-if” challenges in real time.
At the end of a unit	Compare Day 1 model to final model. The growth IS the assessment — students see how their thinking evolved.

5 Slide-by-Slide Facilitation

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Slide 1: Cover

2 min

Say: *"Have you ever walked on hot sidewalk in bare feet? OUCH! But the grass in the shade is cool. Why is that?"*

Watch For

Show two pictures: a sunny sidewalk and a shady spot. Ask students which they would rather sit on during a hot day and why.

Slide 2: What Will We Learn?

3 min

Say: *"Today we will learn how the sun warms things up! We will discover why sunny spots are warm and shady spots are cool."*

Watch For

Introduce the words 'sunlight' and 'temperature.' Practice together. Do a gesture: hands spread wide for sun, arms crossed for shade.

Slide 3: Let's Be Scientists!

3 min

Say: *"Our big question is: Does the sun warm a sunny spot and a shady spot the same? What do you think?"*

Watch For

Quick vote: raise your hand if you think they are the same. Raise your hand if you think they are different.

Slide 4: What Are the Parts?

5 min

Say: *"Our model has two parts. Amount of Sunlight is how much sun shines on a spot — we can CHOOSE a sunny spot or a shady spot. Surface Temperature is how warm it feels — that changes on its own!"*

Watch For

Use sorting mats. Students place Amount of Sunlight on OUTSIDE and Surface Temperature on INSIDE.

Slide 5: How Are They Connected?

5 min

Say: *"When there is MORE sunlight, the temperature goes UP — it gets warmer! That is a POSITIVE connection. When there is LESS sunlight, temperature goes DOWN."*

Watch For

If possible, take students outside to touch a sunny spot and a shady spot on the ground. If indoors, use a lamp on dark paper vs. shaded paper.

Slide 6: What Did We Find Out? 5 min

Say: *"The sun does NOT warm everything the same! Sunny spots are warmer because the sun's energy heats them up. Shady spots stay cooler because something blocks the light!"*

Watch For

Chant: 'When Sunlight goes up, Temperature goes up!' Students complete the sentence frame together.

Slide 7: STEM Challenge 10 min

Say: *"The toy animals are too hot! Can you build a shelter that blocks the sun and keeps them cool? Be creative engineers!"*

Watch For

Distribute cardboard, fabric scraps, craft sticks, tape, and toy animals. Groups build shelters. Test by placing under a lamp or outside in the sun.

6 Answer Key & Expected Responses

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Component Sorting

- **EXTERNAL:** Amount of Sunlight
- **INTERNAL:** Surface Temperature

Relationships

- **Amount of Sunlight to Surface Temperature:** POSITIVE (+) — When more sunlight shines on a spot, the surface gets warmer because light energy turns into heat energy.

Key Discoveries

- Sunny spots feel warmer than shady spots
- The sun heats up the ground, sidewalks, and everything it shines on
- Shade blocks sunlight, which is why shady spots stay cooler
- Dark-colored things in the sun feel hotter than light-colored things

7 Collaborative Learning Protocols

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Why This Matters

ModelIt! is a digital tool, but the deepest learning happens between humans. Computational modeling is most powerful when students build understanding *together* — debating relationships, challenging predictions, and constructing explanations as a community. These protocols ensure every student's voice is part of the model. Use at least ONE per lesson.

Model Gallery Walk

When: After students build models (LEVER Phase V)
Setup: Groups post their models on screens/desks. Half the class stays to "defend," half rotates with sticky notes.
Protocol: Visitors leave 2 sticky notes: one "I notice..." and one "I wonder..." After rotation, defenders read feedback aloud. Class discusses patterns.
Why it works: Students see that the SAME phenomenon can be modeled differently — and that's how real science works. (Bandura: learning through observation of peers)

Scientist Fishbowl

When: During simulation discussion (LEVER Phase E)
Setup: 4-5 students sit in center circle. Rest of class observes with clipboards.
Protocol: Inner circle discusses: "What happened in Scenario 3 and WHY?" using only evidence from their models. Outer circle records strongest arguments. Swap groups.
Why it works: Students practice scientific discourse without teacher directing. Observers learn to evaluate arguments. (Freire: dialogue as the foundation of critical consciousness)

Component Jigsaw

When: During component analysis (LEVER Phase L)
Setup: Assign each group ONE component to become "experts" on. They research it, discuss it, draw it.
Protocol: Regroup so each new team has one expert per component. Experts teach their component to the team. Team builds the model together using everyone's expertise.
Why it works: Every student is essential — the model literally cannot be built without each person's knowledge. Interdependence drives engagement. (Lave & Wenger: learning through legitimate peripheral participation)

Prediction Debate

When: Before running simulations (LEVER Phase E)
Setup: Post a scenario. Students physically move to corners: "Fire Spread will INCREASE a lot" / "INCREASE a little" / "STAY the same" / "DECREASE."
Protocol: Each corner group has 2 minutes to build their argument using model logic. One spokesperson presents. Then RUN the simulation. Losing corners explain what they missed.
Why it works: Movement + social risk + immediate feedback. Students CARE about the outcome because they committed publicly. Wrong predictions become the best learning moments.

Pair Programming for Models (Digital Collaboration)

When: Throughout model building
Setup: Two students, one device. **Driver** controls the mouse/keyboard. **Navigator** directs strategy and catches errors. Switch every 5 minutes.
Protocol: Navigator must verbalize every decision: "I think we should add a negative arrow from Rainfall to Dry Vegetation because..." Driver cannot act without Navigator's reasoning.
Why it works: Forces articulation of scientific thinking. The model becomes a shared artifact, not an individual assignment. Students who struggle with writing can demonstrate understanding verbally. Borrowed from computer science — where the best code is written in pairs.

Quick Protocol Selection Guide

Short on time (5 min)	Prediction Debate — gets bodies moving, generates energy fast
Full period available	Component Jigsaw — deep collaboration, every student contributes

After simulation day	Model Gallery Walk — peer feedback, multiple perspectives
Advanced discussion	Scientist Fishbowl — builds academic discourse skills
Every day default	Pair Programming — transforms individual work into dialogue

8 Differentiation & Extensions

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Below Grade Level

- Pre-sort components as a whole class
- Provide sentence frames for relationships
- Use visual vocabulary cards
- Pair with a strong reader

Advanced Learners

- Add 2-3 additional components
- Research real-world data
- Create multi-step causal chains
- Compare to similar systems globally

English Language Learners

- Pre-teach key vocabulary
- Provide bilingual glossary
- Use visual supports with text
- Allow verbal responses

Gifted & Talented

- Create feedback loop diagrams
- Research prevention strategies
- Write a policy brief
- Connect to climate change

9 Common Misconceptions

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Why This Matters

Addressing misconceptions proactively helps students build accurate scientific understanding. Use these as formative assessment checkpoints during the lesson.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"The sun only warms the air, not the ground"	Sunlight directly warms surfaces it touches — soil, sidewalks, rocks, and water. These warm surfaces then heat up the air above them. The ground can actually be hotter than the air on a sunny day!

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"Shade means it is cold"	Shade does not make things cold — it just keeps them cooler than the sunny spots. On a hot day, the shade is still warm, just not as warm as the sun. Shade blocks the extra heat from sunlight.